Pareto ladder (tightened)

How hard can we compress the LatentMAS relay?

Qwen3-14B · n=100 · seeds 42 & 123 (sample_items) · bootstrap CIs

One upstream build → plain eviction at each B → Judger decode
Greedy · no ASC (memory × accuracy)

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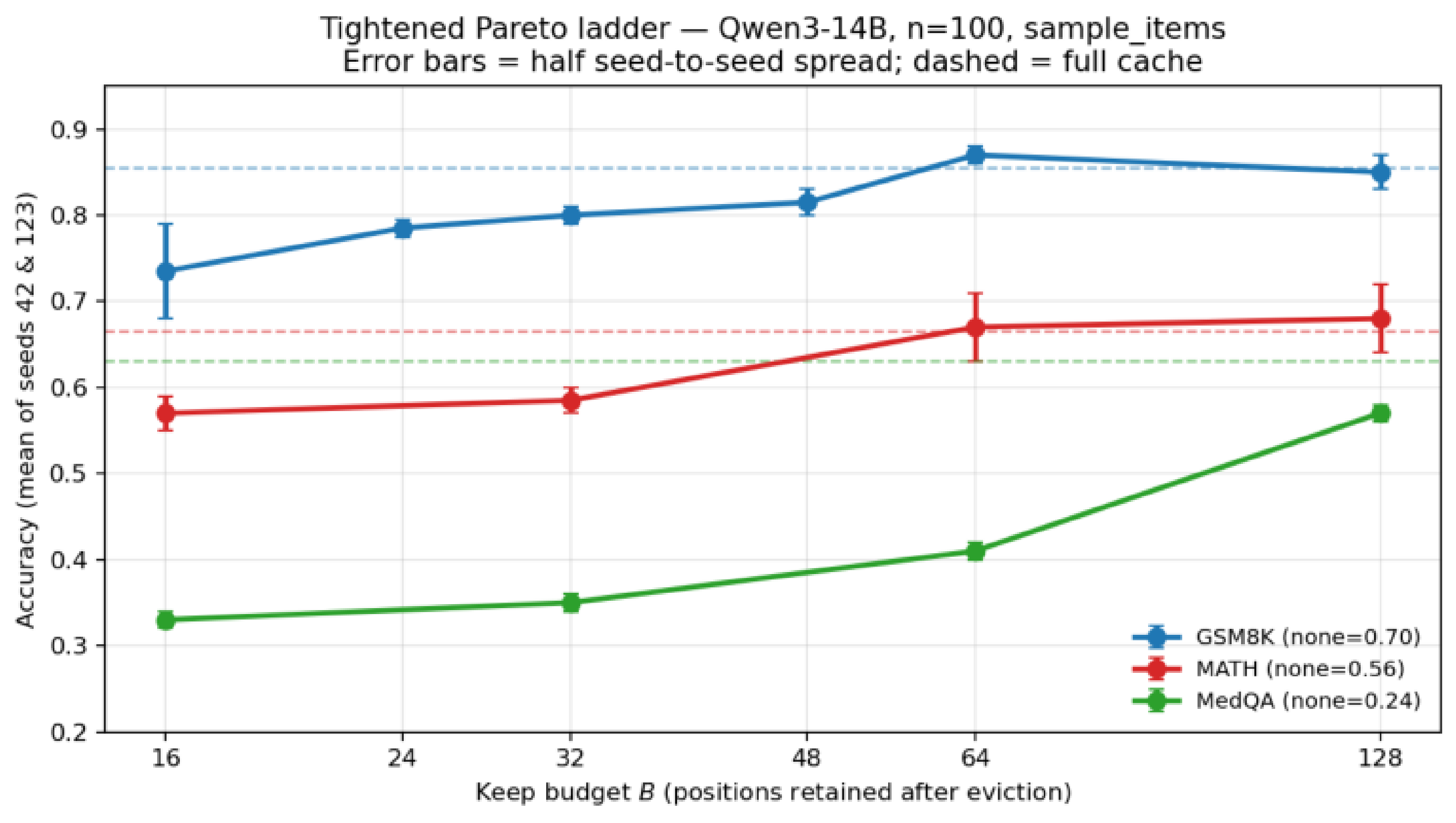
What we are claiming

- LatentMAS relay KV is largely content-insensitive → redundant scaffold
- Systems consequence: shrink it before the Judger (sink + key-norm top-k)
- Question: for each task, how small can B get before accuracy drops?
- This deck: n=100, two real item-subsets, CIs — not the old n=60 identical seeds

Paper-shaped claim

A task-conditional memory–accuracy Pareto — not “40× free everywhere.”

Headline curves (seed-mean ± half-spread)



- GSM8K: soft at B=16; clean iso by B≈64 (~10×); B=32 still mild tax
- MATH: free-ish from B≈64–128 (~5–10×)
- MedQA: still −6 pp at B=128 — thickness required; none collapses hard

Numbers (accuracy, seed-mean n=100)

Task	full	B16	B32	B64	B128	none
GSM8K	0.855	0.735	0.800	0.870	0.850	0.700
MATH	0.665	0.570	0.585	0.670	0.680	0.555
MedQA	0.630	0.330	0.350	0.410	0.570	0.240

Operating points

- GSM8K: B=64 (~10×) for iso-acc
- MATH: B=64–128 (~5–10×)
- MedQA: ≥128; 16–64 collapse

Presence still matters

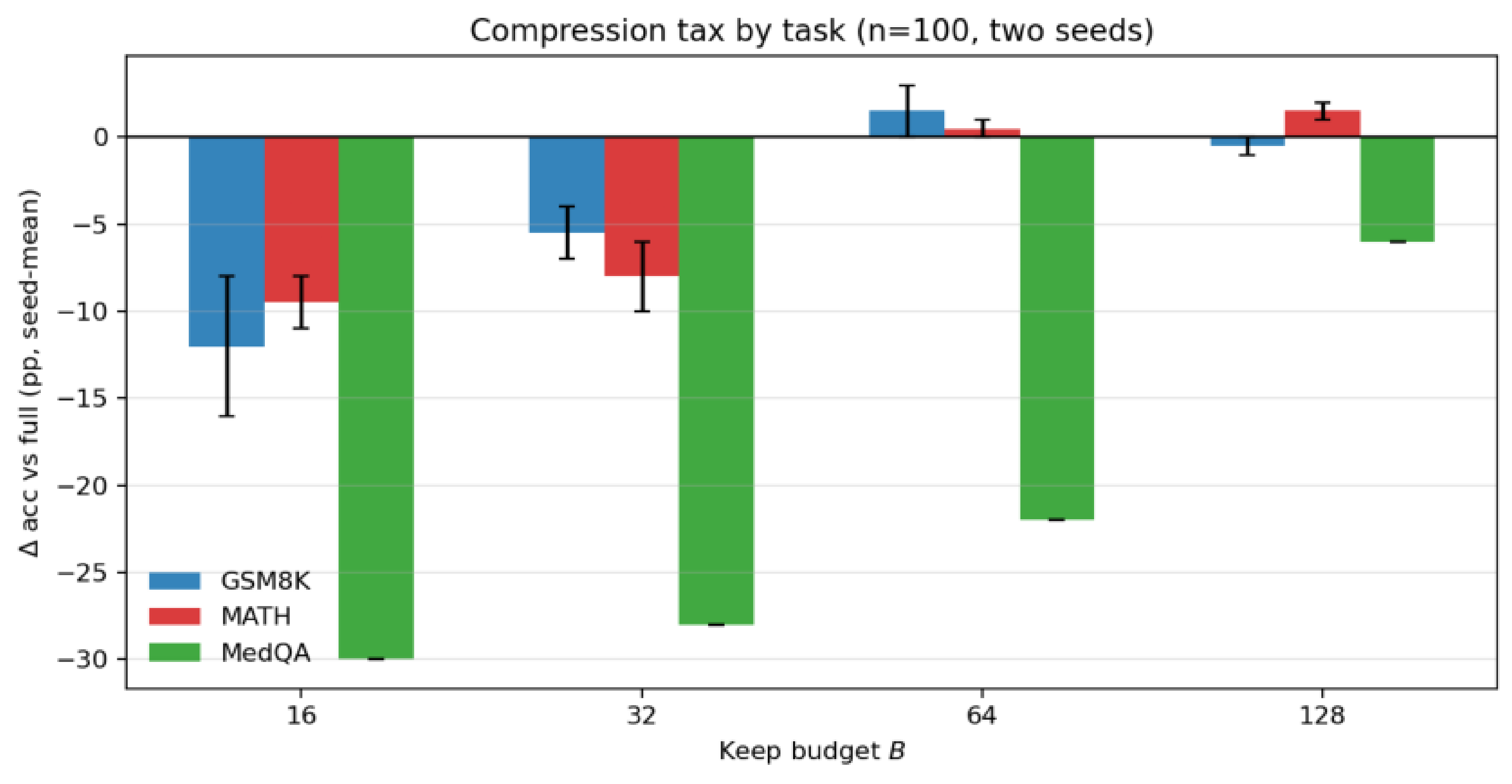
none ≪ full on every task.
Compression trims redundancy;
it does not delete the scaffold.

Hardening: seeds actually differ now

- Old n=60 “two seeds” matched cell-for-cell (greedy + same first-n items)
- Tightened: --sample_items draws different n=100 subsets per seed
- GSM8K B16: 0.79 (s42) vs 0.68 (s123) — variance is real; B64 both 0.87
- MATH shifts with subset; MedQA pool=100 so seeds identical (full set)

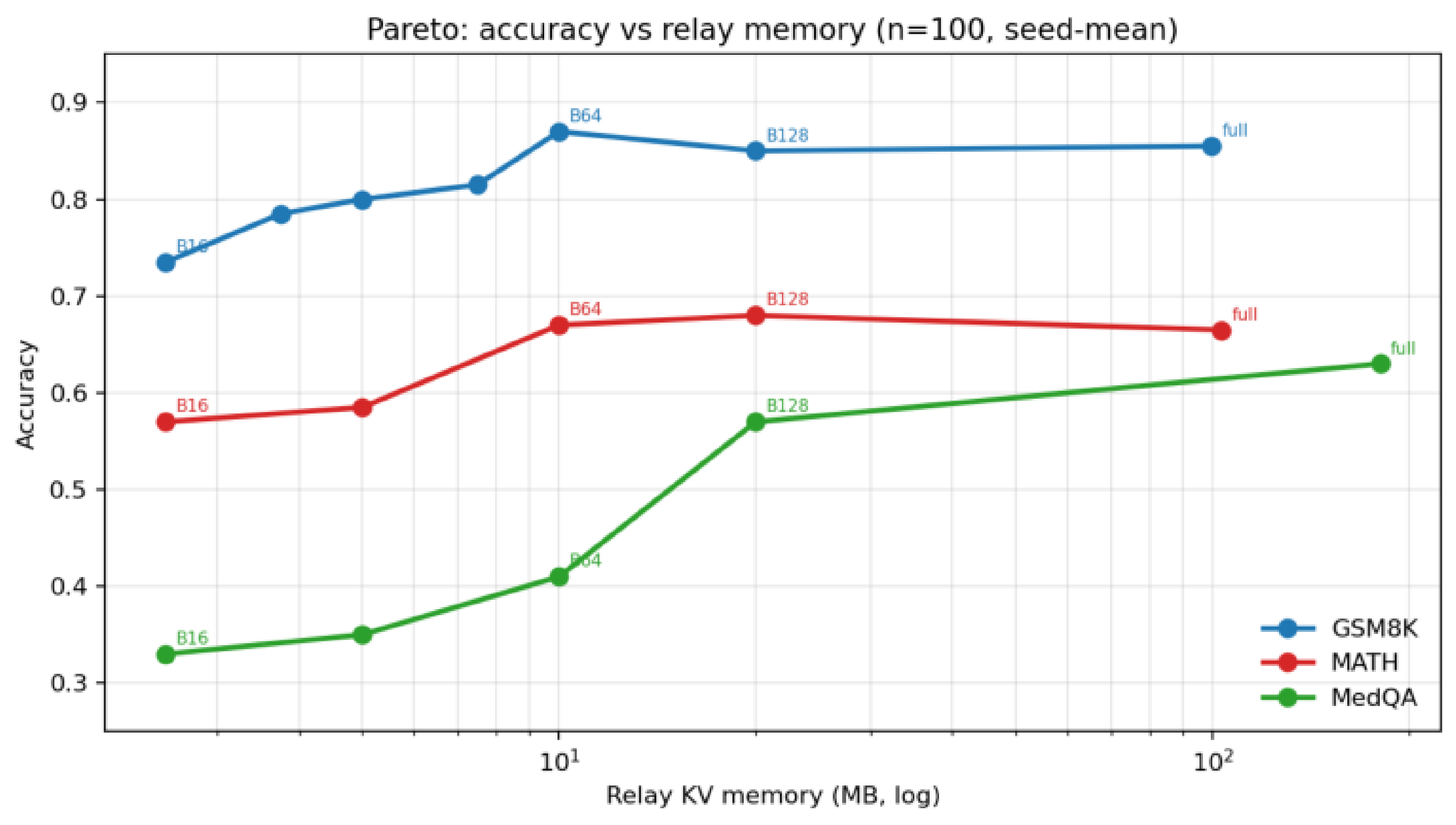
Operating points hold: GSM8K ~64 · MATH ~64–128 · MedQA not free ≤128

Compression tax (Δ acc vs full)



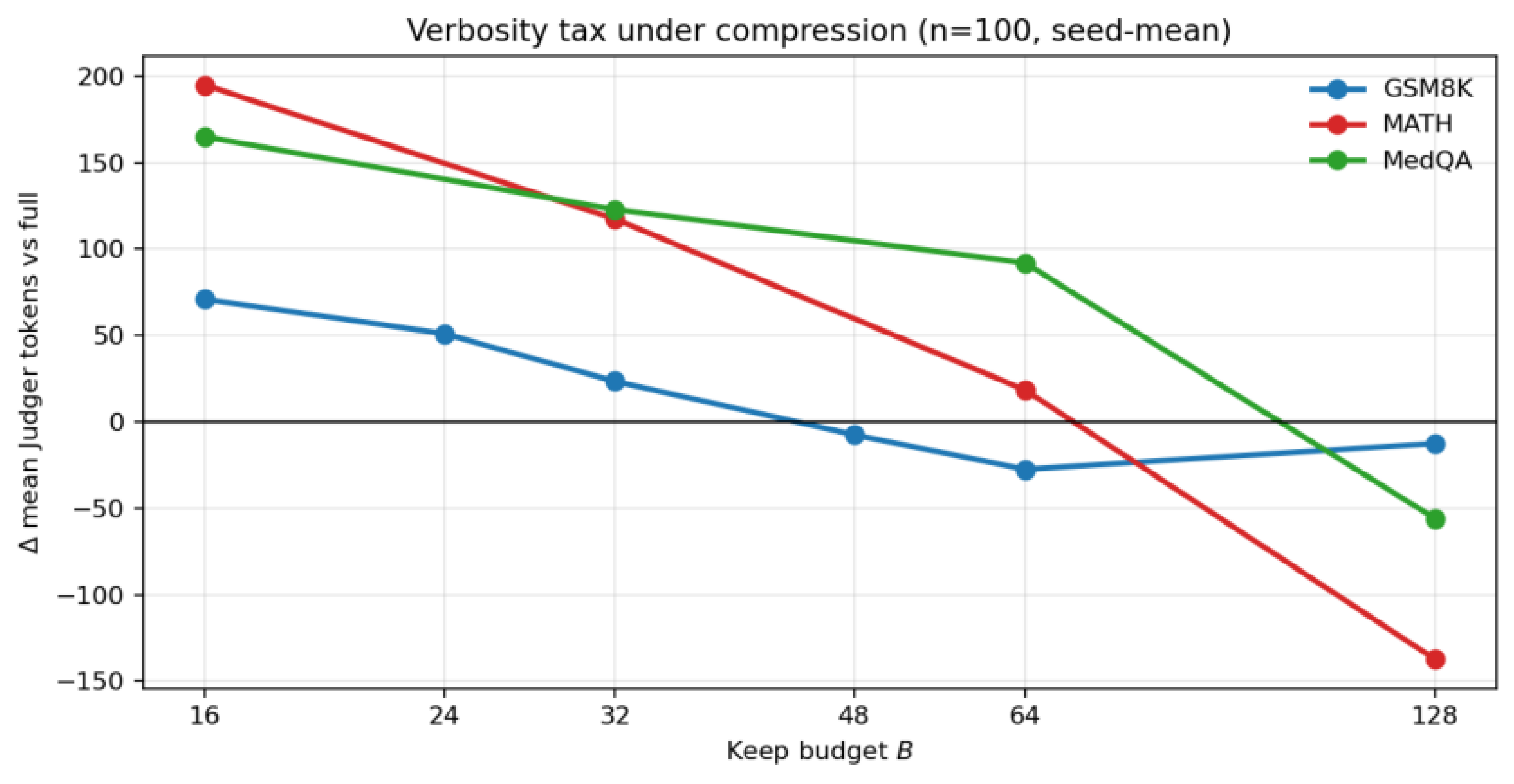
- Easy task ⇒ flat tax until extreme B
- Harder tasks ⇒ tax grows as B shrinks
- Same compressor, different residual — task-conditional keep

Accuracy vs relay memory



- Full relays: ~100 MB (GSM8K/MATH) vs ~180 MB (MedQA)
- B=16 → 2.5 MB (~40–70×); only GSM8K approaches the frontier later
- Sell operating points / Pareto, not one universal ratio

Verbosity tax (Judger tokens)



- Aggressive shrink lengthens Judger CoT (esp. MATH)
- Token tax fades as B grows
- ASC remains the orthogonal length knob at chosen operating points

Bottom line

- Mechanism: relay is a scaffold, not an instance-specific channel
- Systems: compress hard on easy tasks; keep thicker residual on hard ones
- Tightened evidence: n=100, two real subsets, CIs — story unchanged, sharper knees
- GSM8K iso @ B≈64 (~10×); MATH @ 64–128; MedQA still taxed at 128

The scaffold compresses — but not uniformly.
That is the talk.